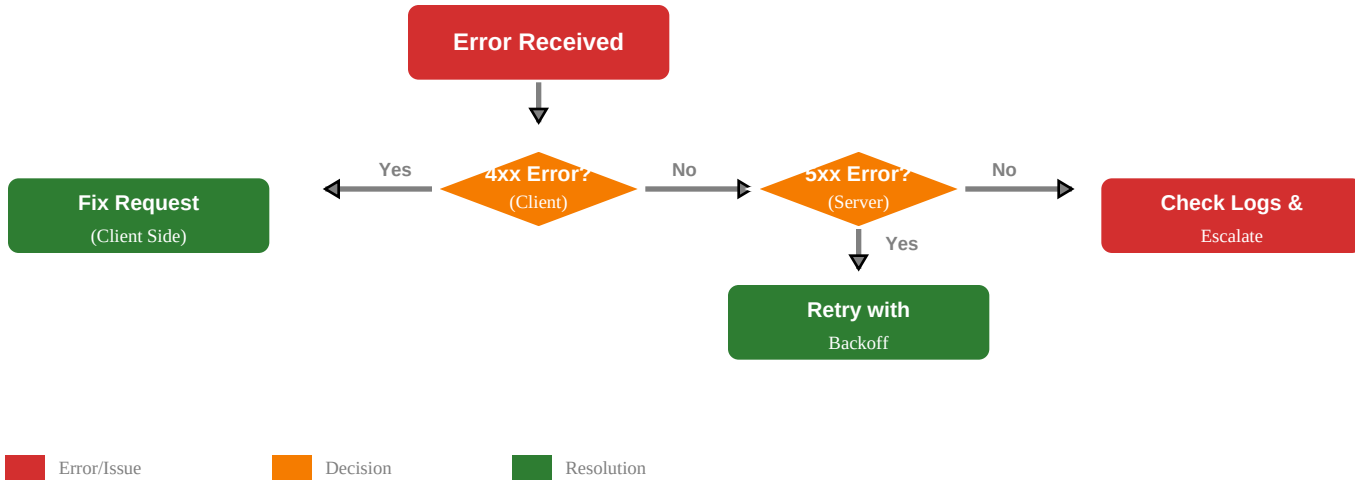


API Error Codes & Troubleshooting Handbook

Enterprise Resolution Intelligence System API v3.5

Error Resolution Decision Flow



1. Authentication & Authorization Failures

Code	Error Type	Cause	Solution
401	Missing Auth	No API key or token provided	Include X-API-Key header or Bearer token
401	Invalid Key	Malformed or expired API key	Generate new API key from dashboard
401	Token Expired	JWT or OAuth token expired	Refresh token or re-authenticate
403	Insufficient Scope	API key lacks required permissions	Request scope upgrade or use correct key
403	IP Restricted	Request from unauthorized IP	Whitelist IP in API settings
403	Account Suspended	Account payment or policy violation	Contact billing or compliance team

Diagnostic Commands:

```
# Test API key validity
curl -I -H "X-API-Key: your_key" https://api.example.com/v3/auth/validate
```

2. Rate Limiting & Throttling

Code	Scenario	Response Headers	Recovery Action
429	Rate limit exceeded	X-RateLimit-Remaining: 0 Retry-After: 60	Wait 60s, implement exponential backoff
429	Burst limit hit	X-Burst-Limit-Reset: 1678...	Reduce concurrent requests
429	Daily quota exhausted	X-Quota-Remaining: 0	Upgrade plan or wait 24 hours

Retry Logic Example (Python):

```
import time
```

```
for attempt in range(3):
    response = requests.get(url, headers=headers)
    if response.status_code == 429:
        retry_after = int(response.headers.get('Retry-After', 60))
        time.sleep(retry_after)
    else: break
```

3. Input Validation & Request Formatting

Code	Issue	Example Error	Fix
400	Invalid JSON	Unexpected token at position 45	Validate JSON syntax
400	Missing required field	Field 'email' is required	Include all required fields
422	Type mismatch	Expected integer, got string	Check data types
422	Value out of range	priority must be 1-5	Use valid value ranges
422	Invalid format	email format invalid	Match expected format/regex
413	Payload too large	Max size: 5MB	Reduce payload or use multipart

4. Server-Side Errors (5xx)

Code	Error	Meaning	Action
500	Internal Server Error	Unhandled exception on server	Retry once; if persists, report with request_id
502	Bad Gateway	Upstream service down	Retry with exponential backoff (2, 4, 8s)
503	Service Unavailable	Maintenance or overload	Check status page, retry after 5min
504	Gateway Timeout	Request exceeded timeout (30s)	Optimize query or increase client timeout

5. Logging & Diagnostic Information

5.1 Essential Headers to Log

- X-Request-ID: Unique identifier for tracing (e.g., req_abc123xyz)
- X-Response-Time: Server processing time in milliseconds
- X-RateLimit- *: Rate limit status headers
- Retry-After: Seconds to wait before retrying

5.2 Error Response Structure

```
{
  "error": {
    "type": "validation_error",
    "message": "Request validation failed",
    "details": [
      { "field": "priority", "error": "must be between 1 and 5" }
    ],
    "request_id": "req_abc123",
    "timestamp": "2026-03-05T10:30:00Z"
  }
}
```

5.3 What to Log for Debugging

- Full request URL and method
- Request headers (excluding sensitive auth tokens)
- Request body (sanitize PII/credentials)
- Response status code and headers
- Response body (full error object)
- Client timestamp and server timestamp

6. Escalation Path

Error Category	Self-Service Actions	Escalate If...	Contact
4xx Client Errors	• Review docs • Fix request format • Check API key scopes	Error persists after fixes	Community forum or developer support
Rate Limiting	• Implement backoff • Optimize requests • Cache responses	Need quota increase	sales@example.com for plan upgrade

5xx Server Errors	• Retry 2-3 times • Check status page • Wait 5 minutes	Errors continue >15min	support@example.com with request_id
Auth Issues	• Regenerate API key • Check IP whitelist • Verify account status	Can't generate new key or account suspended	billing@example.com or support
Critical Production	N/A - immediate escalation	Revenue-impacting outage	emergency@example.com (Enterprise only)

7. Quick Troubleshooting Checklist

✓	Check	Action
■	API Key Valid	Test with /v3/auth/validate
■	Request Format	Validate JSON, check Content-Type header
■	Rate Limits	Check X-RateLimit headers in last response
■	Network Issues	Test with curl, check DNS resolution
■	Logs Captured	Save request_id, full request/response
■	Status Page	Visit https://status.example.com

Support Resources

API Status: <https://status.example.com> | Documentation: <https://docs.example.com>
Community Forum: <https://community.example.com> | Support: support@example.com
Emergency Hotline (Enterprise): +1-800-API-HELP